

Auteur: Xander De Blauwer

Studierichting: Informatica

Oefeningenreeks 7A nr. 14.2

$$f(x, y) = \sqrt{50 - x^2 - y^2} \text{ in } \mathbf{a}(4, -3)$$

$$f(4, -3) = 5$$

$$\frac{\partial f}{\partial x}(x, y) = \frac{-x}{\sqrt{50 - x^2 - y^2}} = \frac{-4}{5}$$

$$\frac{\partial f}{\partial y}(x, y) = \frac{-y}{\sqrt{50 - x^2 - y^2}} = \frac{3}{5}$$

$$f(x, y) = \sqrt{50 - x^2 - y^2} \text{ in } \mathbf{a}(4, -3) = \frac{3}{5}$$

$$z - 5 = \frac{-4}{5}(x - 4) + \frac{3}{5}(y + 3)$$

$$z = \frac{-4}{5}x + \frac{3}{5}y + 10$$

## References